

Artificial Intelligence 2: Bayesian Networks – Exact Inference

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Outline of Topics

- 1 Introduction to Bayesian Network Inference
- 2 Exact Inference

Mathematical Notations

We are using the following shorthand mathematical notations for joint probability distributions of two random variables X and Y using notations of probability:

- The marginal PMF of X : $P(X)$
- The marginal probability of X takes the value x : $p(x)$.
- Joint PMF: $P(X, Y)$
- The value of joint PMF $p(X, Y)$ at (x, y) , i.e., $P_{XY}(X = x, Y = y)$: $p(x, y)$
- Conditional PMF: $P(X|Y)$
- The value of conditional PMF $p(X|Y)$ at (x, y) , i.e., $P_{X|Y}(X = x, Y = y)$: $p(x|y)$

Inference in Bayesian networks

Inference:

- Diagnosis: $P(\text{Cause}|\text{Effect})$, e.g., $P(R|W)$ (See Week 9 Lecture 2, page 14)
- Prediction: $P(\text{Effect}|\text{Cause})$, .e.g, $P(W|C)$ (Discuss in this lecture)

Applications:

- Classification: $\max_{\text{class}} P(\text{class}|\text{data})$
- Decision Marking: $\sum_i P(\text{Effect}_i|\text{Cause}) \times U(\text{Effect}_i|\text{Cause})$

Formally, we divide random variables (nodes) of a Bayesian network into three classes:

- | | | |
|--------------------------|------------------------------------|-----------------------|
| - Evidence variables | $\mathbf{E} = \{E_1, \dots, E_m\}$ | what you know |
| - Query variables | $\mathbf{X} = \{X_1, \dots, X_n\}$ | what you want to know |
| - Non-evidence variables | $\mathbf{Y} = \{Y_1, \dots, Y_k\}$ | what don't care about |

The complete set of variables (nodes) Bayesian network is defined as $\{\mathbf{E} \cup \mathbf{X} \cup \mathbf{Y}\}$

Inference in Bayesian networks

Inference (aka. Query): answering questions about the underlying probability distribution:

- **Unconditional or conditional probability inference:**
 - What is the probability of a given value assignment for a subset of variables $\mathbf{X}$? e.g., $P(W = 1)$
 - What is the probability of different value assignments for query variables $\mathbf{X}$ given evidence about variables $\mathbf{E}$? I.e. compute $P(\mathbf{X}|\mathbf{E} = \mathbf{e})$. e.g., $P(W|C = 1)$
- **Maximum a posteriori (MAP) inference:** Given evidence $\mathbf{E} = \mathbf{e}$, find the most likely assignment of values to the query variables $\mathbf{X}$:

$$\text{MAP}(\mathbf{X}|\mathbf{E} = \mathbf{e}) = \underset{\mathbf{x}}{\operatorname{argmax}} P(\mathbf{X} = \mathbf{x}|\mathbf{E} = \mathbf{e})$$

- Example: Classification, i.e., given the training data and a new example, we want to determine the most probable class label of the new example – $\mathbf{e}$: data and $\mathbf{x}$: label

Exact Inference in Bayesian networks

Exact Inferences in Bayesian networks: to compute the posterior probability distribution of the query given the evidence:

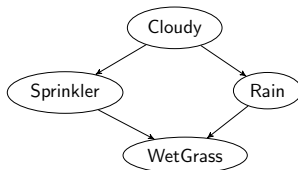
$$P(\mathbf{X}|\mathbf{E}) = \frac{P(\mathbf{X}, \mathbf{E})}{P(\mathbf{E})} = \alpha P(\mathbf{X}, \mathbf{E}) = \alpha \underbrace{\sum_{\mathbf{Y}} P(\mathbf{X}, \mathbf{E}, \mathbf{Y})}_{\text{Marginalisation}}, \quad (1)$$

where $\alpha = \frac{1}{P(\mathbf{E})}$

Bayesian Networks: Wet grass example

$$\frac{P(C=0)}{0.4} \quad \frac{P(C=1)}{0.6}$$

C	$P(S=0)$	$P(S=1)$
0	0.5	0.5
1	0.9	0.1



C	$P(R=0)$	$P(R=1)$
0	0.8	0.2
1	0.2	0.8

S	R	$P(W=0)$	$P(W=1)$
0	0	1.0	0.0
1	0	0.1	0.9
0	1	0.1	0.9
1	1	0.01	0.99

Figure 1: Wet grass example.

Bayesian Networks: Wet grass example

From the Bayesian network, we have:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Parents}(X_i)) \quad (2)$$

$$= P(C)P(S|C)P(R|C)P(W|S, R) \quad (3)$$

Question: Based on this Bayesian network, given it is cloudy, what's the probability that the grass is wet?

Bayesian Networks: Wet grass example

Question: Based on this Bayesian network, given it is cloudy, what's the probability that the grass is wet?

Answer: The question requests the conditional probability $P(W = 1|C = 1)$. To compute the conditional probability, we use the joint probability:

$$P(W = 1|C = 1) = \frac{P(W = 1, C = 1)}{P(C = 1)}, \quad (4)$$

where $P(C = 1) = 0.6$. To calculate the numerator, we need to express it as a marginal distribution

$$\begin{aligned} P(W = 1, C = 1) &= \sum_S \sum_R P(W = 1, S, R, C = 1) \\ &= \sum_S \sum_R P(C = 1)P(S|C = 1)P(R|C = 1)P(W = 1|S, R) \end{aligned}$$

More on marginalisation

Marginal distribution: a distribution formed by calculating the subset of a larger probability distribution of a collection of random variables.

Example: two random variables X and Y with joint PMF and marginal PMFs given in Table

	$Y = 0$	$Y = 1$	$Y = 2$	$P_X(x)$
$X = 0$	$\frac{4}{24}$	$\frac{6}{24}$	$\frac{3}{24}$	$\frac{13}{24}$
$X = 1$	$\frac{3}{24}$	$\frac{4}{24}$	$\frac{4}{24}$	$\frac{11}{24}$
$P_Y(y)$	$\frac{7}{24}$	$\frac{10}{24}$	$\frac{7}{24}$	$\frac{24}{24}$

Table 1: Joint PMF of X and Y and their marginal PMFs $P_X(x)$ and $P_Y(y)$

Calculation: **For discrete random variables we sum**

More on marginalisation

Calculation: **For continuous random variables we integrate.**

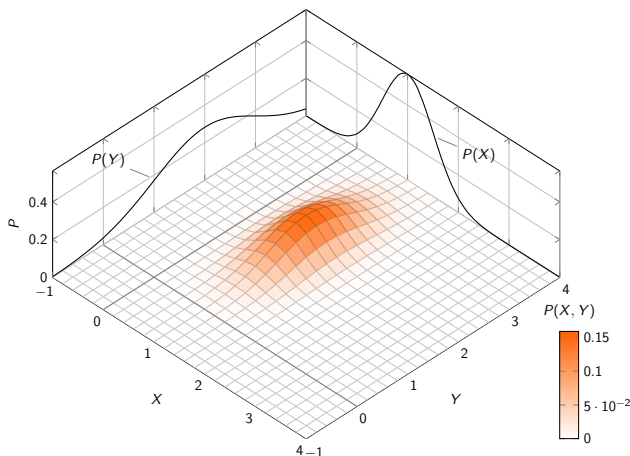


Figure 2: A bivariate normal distribution $P(X, Y)$ and its marginal distributions $P(X)$ and $P(Y)$. Source: [StackExchange: Tex](#)

Bayesian Networks: Wet grass example

(**Continue**) $P(C = 1)$ has been cancelled from the numerator and denominator, we have:

$$\begin{aligned}
 P(W = 1|C = 1) &= \sum_S \sum_R P(S|C = 1)P(R|C = 1)P(W = 1|S, R) \\
 &= \sum_S \sum_R \underbrace{P(S|C = 1)}_{\begin{array}{c|cc} C & P(S=0) & P(S=1) \\ \hline 1 & 0.9 & 0.1 \end{array}} \underbrace{P(R|C = 1)}_{\begin{array}{c|cc} C & P(R=0) & P(R=1) \\ \hline 1 & 0.2 & 0.8 \end{array}} \times \underbrace{P(W = 1|S, R)}_{\begin{array}{cc|c} S & R & P(W = 1) \\ \hline 0 & 0 & 0.0 \\ 1 & 0 & 0.9 \\ 0 & 1 & 0.9 \\ 1 & 1 & 0.99 \end{array}} \\
 &= 0.9 \times 0.8 \times 0.9 + 0.1 \times 0.2 \times 0.9 + 0.1 \times 0.8 \times 0.99 \\
 &= 0.7452
 \end{aligned}$$

Exact inference

Exact inference: Bayesian inference algorithms that calculate the exact value of probability $P(X|E)$.

Inference by enumeration: we inferred a posterior probability by marginalisation of joint distribution, i.e., calculate the exact value of probability $P(X|E)$ using equation 1

Problem: the number of terms in the summation is exponential in the non-evidence random variables. The complexity is $O(nm^n)$, where n is the number of non-evidence variables and $m = |R_Y|$ is the number of values each variable can take.

- **Explanation:** Summation of n terms requires $O(n)$ and each term requires $O(m^n)$ multiplications $\Rightarrow O(nm^n)$.
- **Example:** In the Wet Grass example (page 11), if we do not cancel $P(C = 1)$, then the full calculation requires summing 4 terms each have 4 multiplications.

Exact Inference: Beyond enumeration

Please read [Probabilistic Graphical Models \(find the pdf yourself.\)](#) for

- Variable elimination: Chapters 9.2 – 9.3.
- Clustering algorithm: Chapter 10.1.1
- Sum-product algorithm or belief propagation: Chapters 10.2 – 10.3